

Role of Measured Spherical Tip Geometry in Instrumented Nanoindentation at Shallow Depths

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Abstract

High-throughput characterization of thin-film and combinatorial material libraries demands efficient methods for extracting reliable mechanical behavior from nanoindentation data. A promising route is to combine physics-based finite element modeling with machine learning to link measured spherical indentation responses to underlying stress-strain behavior. A necessary first step in this workflow is the development of a forward modeling framework capable of accurately reproducing shallow spherical nanoindentation, where the load-depth response is strongly affected by indenter geometry. At these small scales, real spherical tips often deviate from their nominal radius, leading to error when idealized geometries are assumed in simulation. In this work, 3D profilometry is used to measure the true indenter shape, and the full measured geometry is incorporated directly into finite element simulations of indenter-specimen interaction. Refractory conductive micron-sized wire alloys are used to evaluate the framework across materials with different yielding and hardening behavior. Using the measured tip geometry together with material properties from tension tests reducing load–depth error from more than 20% with a nominal spherical model to under 5% with the scanned tip geometry, across three metals spanning a wide range of yield strength and hardening behavior. The resulting framework provides an accurate and transferable load-depth foundation for future machine-learning-based inverse extraction of stress-strain behavior from spherical nanoindentation data.

1. Introduction

Instrumented nanoindentation is widely used to evaluate the elastic modulus, hardness, and local deformation behavior of materials at very small scales, and is increasingly adopted as the primary mechanical probe in high-throughput characterization of thin-film and combinatorial material libraries [1–5]. Depth-sensing indentation systems continuously record load and depth

and, when combined with a calibrated indenter area function, remove the need for direct imaging of every imprint [1,5]. As a result, nanoindentation has become central for mapping local mechanical response materials, where the indentation volume is often smaller than individual grains and where properties must be linked to local composition, microstructure, and residual stress [6–9].

With the growth of thin-film technologies, combinatorial sputtering, and high-throughput microstructural libraries, nanoindentation is increasingly applied to micron- and sub-micron films, often in spatially resolved, high-throughput formats [7,10–14]. In these regimes, substrate effects, indentation size effects, and shallow penetration depths can strongly influence the measured load–depth curves and complicate the extraction of intrinsic mechanical properties [10–12,15,16]. Consequently, numerical simulations are widely employed alongside experiments to understand the indentation response and to quantify sensitivity to constitutive parameters, geometry, and boundary conditions [11,13,17–21]. In particular, spherical indenters are attractive in these shallow-depth and thin-film regimes because their gradually evolving mean pressure and representative strain can encode rich information about the underlying stress–strain behavior. However, this same sensitivity also makes spherical indentation especially vulnerable to uncertainties in the assumed tip geometry, so accurately representing the actual (imperfect) spherical tip in finite element simulations becomes critical for interpreting such measurements.

FEM remains the most widely used tool for quantitative analysis of instrumented indentation at micro- and nanoscales. FEM naturally handles nonlinear material behavior, evolving contact, residual stress fields, and complex boundary conditions, and has been applied to nanoindentation of bulk materials, thin films, and layered systems [11,13,17,18,20–24]. Because FEM operates at the same continuum scale as the experimental contact problem, it is particularly well suited for investigating how assumptions about indenter geometry influence the measured load–depth response.

A central challenge in such modeling is the representation of the indenter tip [25]. Most analyses assume a perfect Berkovich or spherical geometry, even though real tips deviate from nominal specifications due to polishing, manufacturing tolerances, and accumulated wear [16,26]. For sharp indenters, these deviations, together with surface roughness and friction, have been shown to affect contact area, hardness, and modulus values, especially at shallow depths

[11,16,26–28]. As a result, a large body of work has developed correction procedures to account for tip rounding, non-ideal tip profiles, and surface imperfections, using analytical relations, finite element supported corrections, or data-driven approaches [12,27–31]. These methods can substantially reduce errors in hardness and modulus over depths from a few hundred nanometers to several micrometers [11,12,16,26–29].

Sharp indenters remain dominant in nanoindentation practice because their steep geometry produces immediate plasticity and allows depth-independent hardness, while the unloading response can be interpreted reliably using the Sneddon/Oliver–Pharr framework [1,4,5]. Within this framework, modest geometric imperfections are often absorbed into the calibrated area function, so the unloading portion of the curve is comparatively insensitive to small errors in tip shape [1,5].

Spherical indenters, in contrast, produce a gradual transition from elastic to plastic deformation with increasing penetration depth and a continuous evolution of mean contact pressure and representative strain [4,7,15,22,32]. This makes the loading curve highly sensitive to yield strength and strain hardening, and has motivated extensive work on extracting stress–strain curves and other elastoplastic parameters from spherical indentation via inverse or reverse-analysis procedures [33–38]. However, the same sensitivity means that inaccuracies in the assumed surface profile or contact geometry can introduce sizeable deviations between experimental and simulated load–depth curves, particularly at shallow depths and near dissimilar interfaces [11,29,39,40]. In many existing spherical-indentation methods, material-specific correction factors or depth-dependent fitting functions are required to reconcile measurements with idealized spherical models such as modified Hertzian contact approaches [25,32,38].

Although many studies acknowledge that real indenter geometry matters, incorporating fully three-dimensional measured tip shapes into FEM nanoindentation models remains uncommon. For sharp pyramidal tips, the geometry evolves rapidly with use, so repeated high-resolution scanning offers limited long-term benefit and most works rely instead on area-function calibration and empirical corrections [16,27,28,40]. Spherical tips, by contrast, are comparatively blunt and mechanically stable: direct measurements have shown that nominally spherical indenters exhibit systematic deviations from the ideal sphere, and that assuming a perfect radius can bias nanoindentation results [11,16,25,31,39]. Because the overall shape of a spherical tip remains

essentially unchanged unless the indenter is damaged, high-resolution three-dimensional scans obtained by profilometry or related techniques can be reused across many experiments, making them both practical and scientifically meaningful [41]. Modern surface-topography methods now make it feasible to reconstruct these geometries with sub-micrometer resolution and to import them directly into finite element meshes [42,43].

The present study is built on this opportunity. Prior work has demonstrated that indenter shape, tip bluntness, surface roughness, and other geometric imperfections can significantly influence nanoindentation responses, and has introduced a variety of correction or compensation strategies [11,12,16,27–31,39–41,44]. However, there has been relatively little systematic analysis of how much improvement is gained, over a broad range of material behaviors, when the measured three-dimensional spherical tip geometry is used directly in forward FEM simulations instead of a nominal radius. In the context of high-throughput nanoindentation of thin-film and combinatorial material libraries, efficient extraction of stress-strain behavior will likely require coupling physics-based forward modeling with machine-learning-based inverse analysis. Inaccurate representation of spherical tip geometry can distort the simulated load-depth response and therefore contaminate the relationship between measured indentation data and the underlying constitutive behavior [41]. Because the load-depth curve serves as the primary observable for inverse analysis, such geometry-induced errors must first be removed at the forward-model level before the resulting data can reliably support machine-learning-based inversion [28]. Establishing the fidelity of the forward simulation with respect to the true indenter geometry is therefore a necessary first step. Here, we establish a reliable foundation for spherical nanoindentation by quantifying how directly incorporating a measured 3D tip shape affects the load–depth response in well-characterized metals with distinct elastic–plastic regimes. Rather than immediately solving the full inverse plasticity problem, we focus here on validating the forward response required for robust physics-based extraction of nanoscale stress-strain curves from spherical indentation [7,8,45–47].

To ensure physical relevance, we compare spherical indentation simulations with experiments and independent tensile tests on three metal wires spanning a wide range of yield strength and hardening behavior. Advances in profilometry make it possible to capture the real spherical tip geometry once with high fidelity and reuse it across all simulations. Embedding this true geometry into FEM provides high-quality, geometry-consistent load–depth data without material-specific

calibration curves or ad hoc correction factors [11,13,17,21,23,44,48]. As shown in this work, using the measured tip shape produces consistent responses across multiple alloys and improves agreement with experiment by more than 10% relative to nominal-radius models, establishing a transferable description of the contact that can be combined with machine-learning-based inverse analysis to recover material properties from spherical indentation data. This forward modeling framework thereby serves as the necessary first step toward a fully integrated physics-based and machine-learning pipeline for high-throughput mechanical characterization of thin-film and combinatorial material libraries.

Because the indenter geometry is a property of the tip rather than the specimen, the same measured shape can be applied unchanged across different materials, loading conditions, and, in future work, thin-film and combinatorial libraries [11–14,16,23,49]. This universality distinguishes the present approach from existing spherical-indentation methods that rely on material-specific and depth-dependent corrections, and provides a general, experimentally grounded starting point for extending spherical nanoindentation to shallow depths where geometric fidelity is essential.

2. Experiment methods for spherical nanoindentation

2.1. Material selection

The focus of this study is to quantify and ameliorate the errors associated with shallow depth spherical indentation where the errors are defined in terms of deviation between the simulated and experimental load–depth curves. Most nanoindentation systems use force-controlled actuators, applying a prescribed load while measuring depth and contact stiffness. In contrast, finite element simulations solve the forward problem: material properties and constitutive laws are inputs, and load–depth curves, pile-up profiles, and contact stiffness responses are outputs. Therefore, FEM simulations require the knowledge of material properties a-priori. From an experimental perspective, we assume isotropic behavior under indentation, which would be an invalid assumption if the indentation scale is sub-granular, as single crystals are typically anisotropic. Therefore, the material chosen for this study must satisfy the following: 1) there should be a way to measure the properties of the material for input to the FEM simulations, 2) the measured properties should reflect the properties at a scale where spherical indentation is conducted, and 3) the grain structure of the material should be such that multiple grains are encountered under

spherical indentation to average out the anisotropy. For this study, we have selected metal wires, specifically copper (Cu), molybdenum (Mo), and tungsten (W). Copper (Cu) is characterized by its 99.95% purity, adhering to PO# E011060078 / EA SO# 29290 with wire chemistry following ASTM-B-387 type 361 standards, and measures about 50 micrometers in diameter. Similarly, molybdenum (Mo) maintains a 99.95% purity with the same PO# and standards, measuring about 50 micrometers in diameter. Tungsten (W) also features wire chemistry specifications suitable for this study, measuring about 50 micrometers in diameter. These materials are selected due to their wide range of yield and ultimate strength properties, which are crucial for validating FEM simulations. It should be noted that Cu wire is the material under primary investigation. Mo and W are used as a source of validation materials for the findings of this work. Figure 1 shows the scanning electron microscopy (SEM) and electron backscattering diffraction (EBSD) scan of Cu wire. The diameter of the wire is $\sim 45 \mu\text{m}$, which is similar to the $\sim 5 \mu\text{m}$ radius of the contacting indenter. Moreover, based on the EBSD scan within $\sim 10 \mu\text{m}$ diameter of contact, the indenter will encounter several grains.

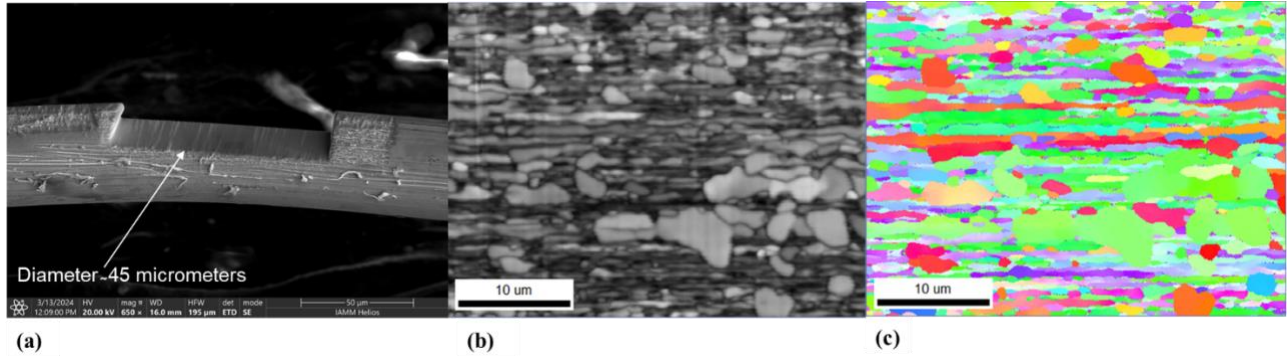


Figure 1. Scanning Electron microscopy (SEM) analysis of copper (Cu) wire, **(a)** SEM image of the cut cross-section of the Cu wire using the secondary electron (SE) detector, **(b)** SEM image of the same region using the backscattered electron (BSE) detector, and **(c)** electron backscatter diffraction (EBSD) map revealing the nanoscale grain structure of Cu wire.

2.2. Tension testing of metal wires

Tension tests were conducted using the Psylotech xTM, equipped with a 1.6 kN load cell and a 0.1 mN resolution. The wires were bonded to Garolite 10 (G10) tabs using epoxy cured by mixing hardener in a 4:1 weight ratio. Sections of G10 measuring 3 mm by 6 mm were cut, and wires were bonded to the center of these sections with the aid of an optical microscope to ensure precise

alignment. The wires, now bonded with G10 tabs at both ends and having a gauge length of 25.4 mm, were placed within L-grips of the Psylotech system that were laterally tightened using grub screws. Engineering strain was calculated by dividing displacement by gauge length, with a strain rate set at 0.002 per minute according to [50]. The true stress-strain relationships were obtained by converting engineering stress and strain using standard relationships given by equations 1 and 2. Figure 2 illustrates the mechanical testing setup and presents the resulting true stress-strain curves. Then, the yield strengths were evaluated from the tensile stress–strain curves using an adaptive offset method, and the values are reported in **Table 1**. Additional details of the procedure can be found in supplemental information.

$$\sigma_{true} = \sigma_{engineering} * (1 + \epsilon_{engineering}) \quad (1)$$

$$\epsilon_{true} = \ln (1 + \epsilon_{engineering}) \quad (2)$$

For forward FEM modeling, the experimentally measured tensile stress-strain data were used as input. The plastic flow behavior was provided to the FEM software Abaqus in a tabulated form under the assumption of isotropic hardening. The table consisted of yield stress values corresponding to incremental plastic strain steps of 0.0001, extracted directly from the true stress-strain curves. Meanwhile, Poisson’s ratios for each material at room temperature were taken from the literature [51–53].

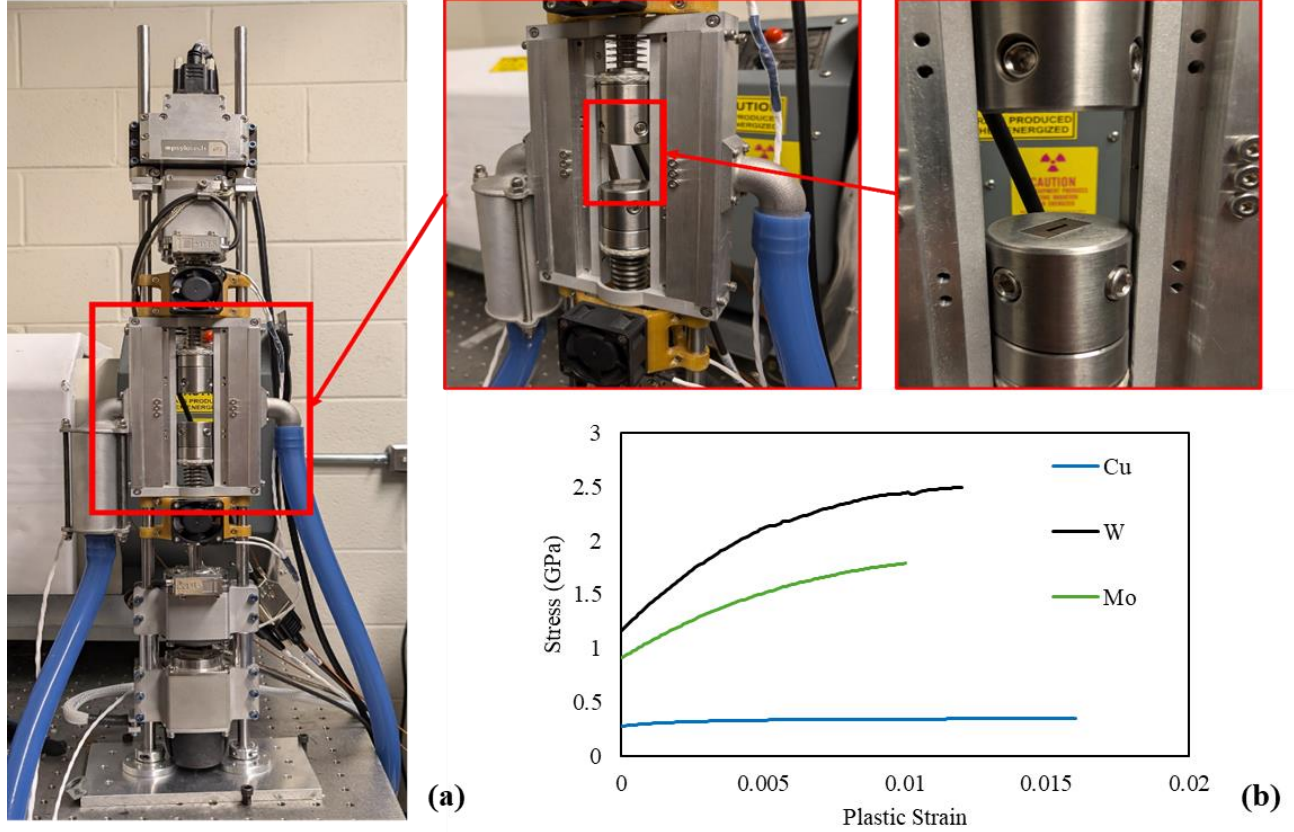


Figure 2. (a) Psylotech tension testing setup with inset showing the gripping fixture, (b) Example true stress strain curve obtained for copper, molybdenum, and tungsten.

Table 1 Tensile tests measured elastoplastic properties of three elements/materials under consideration.

Notation	Young's Modulus, E (GPa)	Poisson's ratio, ν (-)	Yield Strength (MPa)	Ultimate Strength (MPa)
Cu	109 ± 3.2	0.34	269 ± 9.3	339 ± 14.2
W	333 ± 11.6	0.28	1292 ± 249	2630 ± 332
Mo	234 ± 12.2	0.31	967 ± 218	1896 ± 135

* Minimum of 5 samples tested per material type.

* E is obtained from Berkovich Indentation, ν is from literature, Yield Strength and Ultimate Tensile Strength is obtained from Tension Tests

2.3. Experimental nanoindentation using spherical tip

Nanoindentation was done using the iMicro nanoindenter obtained from Nanomechanics Inc. (now KLA). The indenter sits on a vibration isolation table and is installed within an environmental chamber. The samples for indentation are prepared by mounting the wires on an aluminum cube using copper tape. Epoxy is prepared by mixing epoxy cure resin and hardener in a four-by-one ratio, which is then put into a cylindrical mold containing the wires-aluminum cube assembly. Once the epoxy is cured (~8 hours), the samples are taken out of their respective molds and polished using the Buehler handbook guidelines.

One of the considerable challenges associated with the spherical indentation of metal wires is the error resulting from the compliance of wire-epoxy assembly. The compliance can be accounted for, as has been proposed in the literature [54], using equation (3), where it is assumed that the measured stiffness is the combined response of the contact stiffness ($S_{contact}$), the frame stiffness (K_{frame}), and the stiffness of the wire-epoxy assembly ($S_{substrate}$).

$$\frac{1}{S_{measured}} = \frac{1}{S_{contact}} + \frac{1}{S_{substrate}} + \frac{1}{K_{frame}} = \frac{1}{S_{contact}} + \frac{1}{K_{effective}} \quad (3)$$

Therefore, a $K_{effective}$ can be defined, accounting for the substrate stiffness (wire) and the frame stiffness. However, implementing this requires prior knowledge of substrate stiffness. One way to measure substrate stiffness is to load the wire using a flat tip of a slightly smaller diameter than the diameter of the wire and then use the slope of the load-depth curve to evaluate the substrate stiffness [55]. However, testing would require different tip diameters for applications with different shapes and diameters of wire, which would be cumbersome. Moreover, the above correction would correspond to the overall response of the wire and not the local response. Therefore, a much more accurate way of determining the substrate stiffness is by conducting Berkovich indentation near the center of the wires (where the spherical indentation is carried out) and tracking the evolution of Dynamic Stiffness squared divided by Load ($\frac{S_{measured}^2}{P}$) – a variable that remains constant for a material with constant modulus and hardness [5] – as a function of depth. **Figure 3(a)** shows the evolution of $\frac{S_{measured}^2}{P}$ as a function of depth for different values of $K_{effective}$. Note that for the case where $K_{effective} = K_{frame}$, the $\frac{S_{measured}^2}{P}$ decreases significantly as a function of depth. As the value of $K_{effective}$ approaches 9.8e5 N/m, $\frac{S_{measured}^2}{P}$ remains

significantly constant with depth. More importantly, with a slight increase or decrease in $K_{effective}$, $\frac{S_{measured}^2}{P}$ changes significantly, which leads to the determination of $K_{effective}$ sufficient enough to accurately capture the depth – a variable which is weakly dependent upon $K_{effective}$ in comparison to $S_{contact}$. The advantage of the above method is that it is performed at a similar scale compared to the spherical indenter. Therefore, as long as the indentation is performed at/near the center of the wire for both Berkovich and spherical indenter, the substrate effect can be captured. **Figure 3(b)** shows examples of the indentation curve for Cu, Mo and W before and after accounting for frame stiffness. The $K_{effective}$ obtained for Cu, Mo and W are 3.2×10^5 N/m, 6.3×10^5 N/m and 9.7×10^5 N/m respectively. The stiffness is evaluated as the average for 5 Berkovich indentations performed at the center of the wires with a displacement rate of 50 nm/s (same as the displacement rate of spherical indentation). A higher difference between the corrected and uncorrected load–depth curves obtained for W and Mo is expected as the effect of substrate compliance would be low at lower loads (as obtained in Cu indentations) even though the substrate stiffness for Cu is also lower.

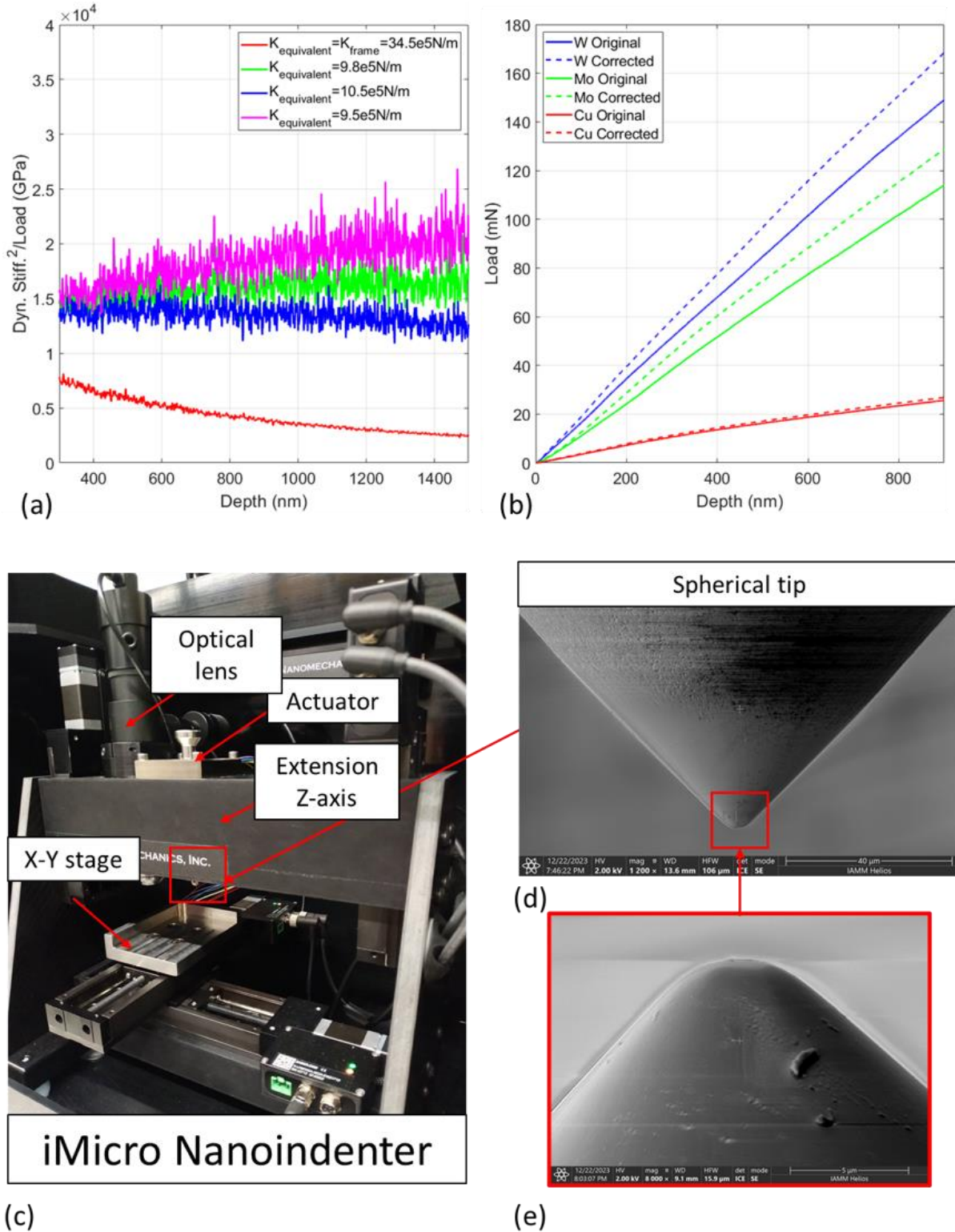


Figure 3. Experimental nanoindentation on copper (Cu) wires using a spherical tip. (a) Equivalent frame stiffness from Berkovich tests. (b) Load–depth curves before and after compliance

correction from spherical tests. (c) iMicro nanoindenter setup. (d) **scanning electron microscopy (SEM) image of spherical tip.** (e) Magnified view of tip apex.

2.4. Laser scanning of actual spherical tip

It is quite common for microscale tips, particularly spherical indenters, to be challenging to acquire due to precise fabrication and calibration difficulties. Minor deviations from the intended shape can lead to significant errors in material property assessments. This is because they are often manufactured by hand, resulting in indenters that are not perfectly symmetrical or spherical. As a result, the load may not be uniformly applied, leading to uneven indentations and skewed results. Furthermore, imperfections in the geometry of the tips significantly affect the model, as the initial input geometry is poorly defined. In the present case study, we procured a spherical indenter whose specifications, as provided by the manufacturer, asserted that one face of the indenter possessed a radius of precisely 4.55 μm , while the opposing face exhibited a variance in radius ranging from 4.60 μm to 4.80 μm . Given that the radius measurements were inadequately detailed, there is an imperative need for a novel methodology capable of capturing the comprehensive geometry of the indenter tip from every conceivable angle.

Minor inaccuracies in the indenter's geometry can significantly affect deformation behavior, particularly at the onset of indentation. Although theoretical models often assume a perfectly symmetrical tip, real-world indenters commonly exhibit subtle deviations that can alter the shape and size of the initial plastic zone. Consequently, these geometric imperfections influence both stress distribution and the magnitude of plastic strain in the substrate, potentially leading to results that differ from idealized predictions [40].

Thus, the 3D geometry of the tip was measured by conducting a laser scan using the VK-x250 laser scanning microscope with a height resolution of 9nm. Due to the high resolution, the microscope performs well in detecting and measuring extremely small vertical deviations on surfaces, which is essential for studies requiring high precision. To prepare the tip, firstly, a SEM stub, shown in Figure 4(a), was customized to the thread specifications of the iMicro nanoindenter to mount the tip. This process ensures that the indenter tip is securely mounted, aligned, and positioned correctly for high-resolution imaging and analysis. Subsequently, the tip now mounted on a SEM stub is then mounted on a SEM sample holder to ensure precise alignment during the laser scan. A double scan at 500x magnification was performed to eliminate any axial-z noise from

the data. The resulting height map was exported as shown in Figure 4(b) and 4(c), and the x-y pixels were accurately scaled. At 500x magnification, the working distance between the lens and the sample is less than 1 mm. Therefore, laser scanning often results in highly saturated blurry images. To account for that, laser scanning was carried out with noise filtering and minimum aperture opening. Following this, the x, y, and z point maps were imported into Civil 3D shown in Figure 4(d) as a comprehensive list of points. A surface was then fitted over these points and subsequently exported as a .SAT file for detailed analysis in Abaqus.

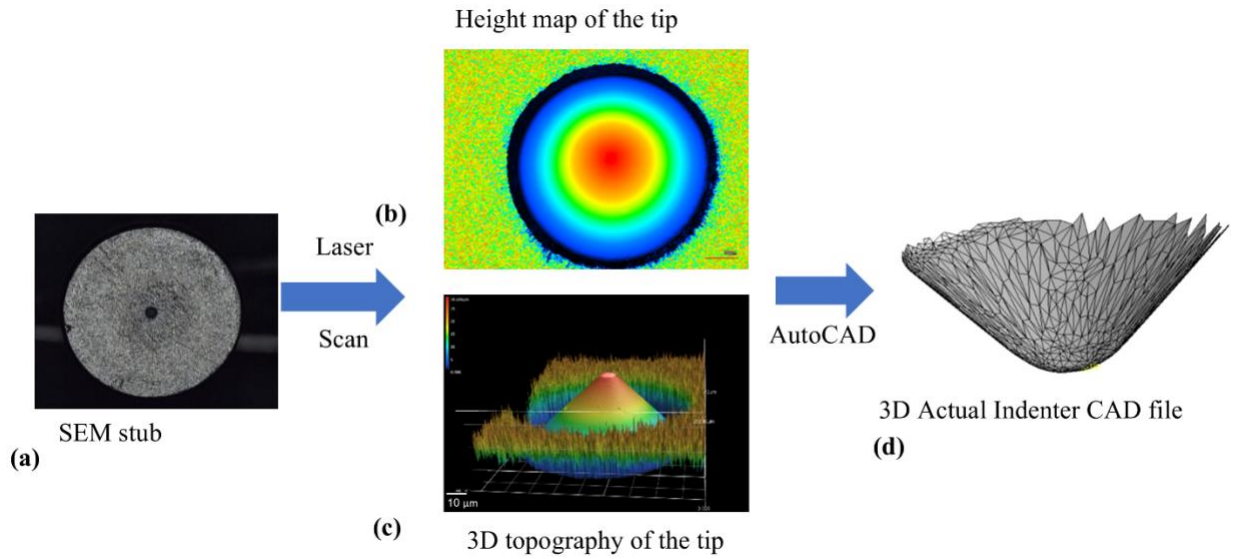


Figure 4. (a) scanning electron microscopy (SEM) stub with threaded hole (matching thread as nano indenter) for tip; **(b)** height map of the tip; **(c)** 3D topography of the tip; **(d)** 3D CAD file.

3. Computational methods for spherical nanoindentation

In the literature, spherical indentation simulations are often conducted using 2D axisymmetric simulations due to their reduced complexity and enhanced computational efficiency compared to 3D representations. However, the current work focuses on evaluating and improving the small-scale imperfections of the actual indenter. Therefore, for a direct comparison between simulations and experiments, a 3D representation of the assembly is necessary.

In this work, the commercially available Abaqus/Standard simulation software is utilized for all simulations. Unless specified, the sample under consideration is a Cu wire with a diameter of 45 micrometers. Consequently, the current study is only interested in the loading portion of the load–depth curve where the depth has been corrected as previously mentioned. The simulation is

performed using a static step, assuming that the experimental indentation is carried out at a displacement rate that is low enough to obtain a static response from the material. The simulation is set for a total time of 2.5 seconds, with a maximum allowed increment of the indenter of 0.01 nm. This increment results in a maximum increment of 1 nm for the total depth of 250 nm. This process ensures that the simulation aligns well with the experimental conditions, leading to a meaningful and comparable result. Three mesh sizes are implemented in the current work: a coarse mesh, medium mesh, and fine mesh, as listed in **Table 2**. **Figure 5** shows the tetrahedral meshes of the specimen at the different resolutions, where the element size is biased near the center of the specimen and larger elements are generated closer to the bounding faces of the specimen using a mapped structure [56]. Henceforth, we refer to these meshes as coarse, medium, and fine mesh, respectively. Unless explicitly mentioned, the contact between the indenter and the sample is assumed to be frictionless. FEM simulations were conducted on a cluster with nodes each having a 20-core CPU and 512GB of memory. The duration of the simulations varied, ranging from several hours to about a day, depending on the mesh size, indentation depth, and the complexity of the model.

Table 2 Comparison of mesh characteristics among coarse, medium, and fine (FEM) simulations

Mesh Type		Element Type	Element Shape	Geometric Order	Number of Elements
Coarse	Specimen	C3D10	tetrahedral	quadratic	37,960
	Indenter	R3D3	triangular	linear	6,424
Medium	Specimen	C3D10	tetrahedral	quadratic	118,070
	Indenter	R3D3	triangular	linear	14,720
Fine	Specimen	C3D10	tetrahedral	quadratic	533,669
	Indenter	R3D3	triangular	linear	42,256

For comparison, a spherical tip with a radius of 4.60 micrometers was selected. This value falls within the range of manufacturing specifications provided by 4.55 to 4.80 micrometers. Figure 5(d) presents the actual geometry of the conical spherical tip in the simulation, which employs 3,833 nodes and 14,720 R3D3 linear triangular elements. **Table 1** provides a description of the elastoplastic deformation of Cu, Mo and W wires, with the plastic behavior of the material depicted in **Figure 2**. Following **Figure 3(b)**, the plastic response was implemented in Abaqus as a

tabulated flow stress versus plastic strain curve (i.e., true stress–equivalent plastic strain input). Plasticity was modeled using the Abaqus isotropic von Mises elastoplastic formulation with equivalent plastic strain as the hardening variable. The parameters utilized in the modified constitutive model presented in **Table 1** are determined based on the conducted tensile tests.

Rate-dependent plasticity and indentation size effects were not explicitly incorporated in the present simulations. The study focuses on shallow spherical indentation under a single controlled loading condition, where deformation remains confined to a limited near-surface volume and strain-rate variations are expected to be secondary relative to the geometric effects under investigation within the shallow-depth loading regime considered here. In addition, the tested materials are nanocrystalline metallic wires, for which prior studies indicate that indentation size dependence is expected to be reduced because the characteristic grain size is already comparable to the deformation length scale at shallow penetration depths [57]. Under these conditions, a rate-independent continuum plasticity model was considered sufficient for establishing the forward geometric fidelity targeted in this work.

The boundary conditions in the simulation were designed to replicate those expected in the experimental tests. Specifically, the bottom of the specimen was constrained with a fixed displacement boundary condition, as these degrees of freedom are similarly constrained in the experimental setup. The indenter was allowed to displace exclusively along the z-axis to mirror the experimental conditions. Given the substantial difference in Young's modulus between the diamond indenter and the specimen, the mechanical properties of the diamond do not influence the load–depth response. Consequently, the indenter was treated as a rigid body in the simulation to simplify the computations, a common assumption in similar studies [29].

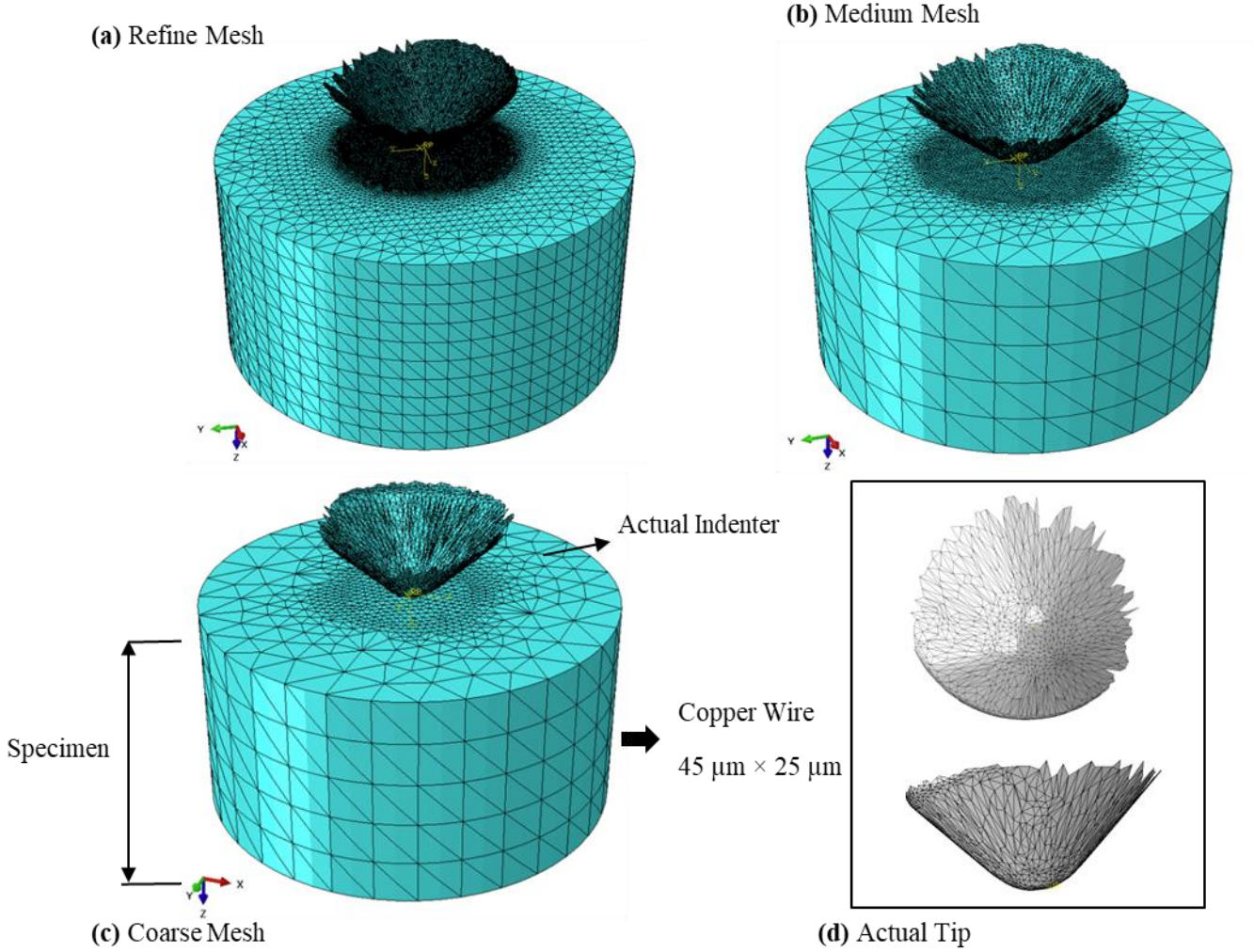


Figure 5. (a) 3D finite element model (FEM) of nanoindentation of pure copper (Cu) with fine mesh; (b) 3D FEM model of nanoindentation of pure Cu with medium mesh; (c) 3D FEM model of nanoindentation of pure Cu wire with coarse mesh; (d) Actual tip

4. Results and discussion

In this section, we describe the procedure used to develop the FEM framework that significantly reduces errors between FEM and nanoindentation tests. We will detail the percentage of error reduction achieved using this method with the actual tip. To accomplish this, several features are extensively investigated, including mesh optimization, friction effects, boundary conditions, and the influence of indenter radius. Our findings indicate that the most critical factor contributing to impressive accuracy compared to other methods is the actual geometry of the tip. The following sections will provide a detailed explanation of these aspects and compare the

individual load–depth curves obtained from both modeling and experimental results conducted on Cu, Mo and W wires.

4.1. Actual tip model

In the FEM simulations, the actual geometry of the indenter was utilized to evaluate the effectiveness of an indenter. This was achieved by comparing the results of FEM models with experimental data obtained from nanoindentation tests conducted as previously described.

Five experimental trials were presented for each alloy to ascertain its characteristics. Subsequently, an average was computed from these tests to facilitate a comparative analysis with simulations employing both actual and spherical tips assuming frictionless contact. Given that the future direction of this research is the examination of shallow depths within thin films, the primary depth for both simulation and experimental observations was established at 250 nm. A comprehensive inquiry into analysis at greater depths will be outlined in the subsequent discussion section of this paper.

First, Figure 6(a) presents a comparative analysis of the load–depth curves obtained from five nanoindentation tests on Cu. The curves are compared to the actual tip model and the spherical model. The “average test” line plots the corresponding FEM simulated load–depth curves generated for Cu wire. These curves exhibit a strong correlation with the mean curve from the experimental data in **Table 3**. The actual simulation curve, when compared to a high-quality experimental test, achieves an approximate accuracy of 98.85%, with an error percentage (defined here as the maximum vertical deviation in load between the simulated and experimental curves at the same displacement over the loading segment) of just 1.15%. This level of accuracy translates to an average error of 0.07 mN between the actual tip model and the experimental test results. If all tests, including the two additional faulty tests, are considered, the error percentage rises to 2.98%. In comparison, simulations utilizing the spherical model exhibit a lower accuracy with an error percentage of 21.21%, indicating that our method of using that actual tip achieves a more precise correlation with empirical measurements than conventional models. The simulation closely follows the trend of the actual nanoindentation test results for Cu samples, indicating that the model accurately predicts the behavior observed during physical tests. The “Spherical Tip” model deviates significantly, showing a different response under nanoindentation. This comparison

highlights the effectiveness of using the actual tip geometry in simulations for replicating experimental data.

Similarly, Figure 6(c) presents the experimental results for W from Tests 1 and 2, which closely align with the simulation curves. This suggests a high level of correspondence between the model and the actual behavior. Conversely, the “Spherical Tip” simulation demonstrates the lowest agreement with the experimental data, highlighting the superior accuracy of the actual tip simulation. **Table 3** presents a comparison of the error percentages between the actual tip model and the average of nanoindentation tests. The actual tip model exhibits an error percentage of 3.32%, significantly lower than the spherical tip model, which has an error percentage of 17.66%. Furthermore, Figure 6(b) presents the load-depth curve results for Mo. From the graph, it can be inferred that, overall, the curve suggests that the actual tip model fits better when compared to the spherical tip model. We find that the mean percentage error for the actual tip model is approximately 6.90%, whereas for the spherical tip it is around 12.81%. The errors could potentially be reduced to less than 6.90% by refining the mesh and smoothing the curves. However, refining the mesh will increase computational costs.

Extending the analysis for Cu up to 500 nm depth, the measured load consolidates near 17–18 mN across all five tests shown in Figure 6(e) with minimal inter-test dispersion, indicating stable experimental repeatability. The simulation with the actual indenter geometry reproduces the average experimental response with high fidelity, showing ~2% error at 500 nm. In contrast, the spherical-tip approximation departs substantially from the data, with ~16% error at the same depth. These results demonstrate that accurate tip geometry remains critical for quantitative agreement at higher penetration up to 500 nm while also confirming the robustness of the experimental curve family. We emphasize this shallow-depth range because tip geometry and surface effects dominate model–experiment differences here, and because future thin-film applications require limiting the indentation depth to avoid substrate influence.

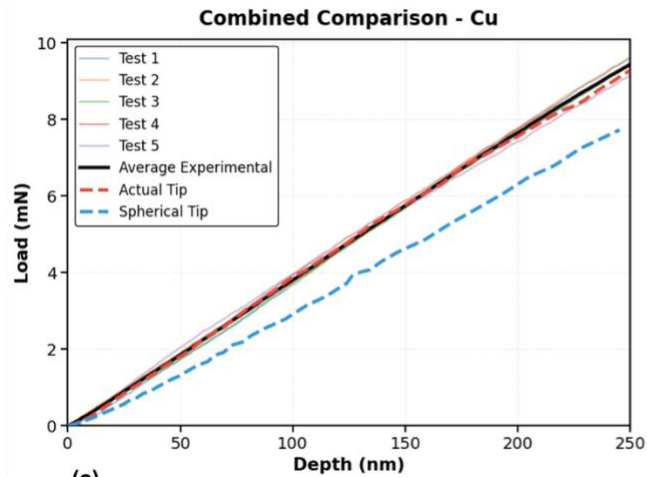
Across all metals, distinct forces are required for indenting the specimen, as evident from the load–depth curves. The models of nanoindentation performed on the 3 alloys exhibit varying degrees of error; however, they effectively represent the load–depth by utilizing the actual tip. Based on these observations and the finite element nanoindentation simulations, several conclusions can be made. Initially, this suggests that, with respect to actual tips, a higher force is

necessary at deeper depths due to their asymmetry. These tips often possess irregular shapes, such as slight flattening on one side, having an edge, or being elliptical rather than perfectly smooth circles. Such geometry potentially leads to variations in contact area shape across the depth of the indenter. Consequently, stress distribution becomes uneven, with certain areas experiencing higher stress concentrations while others bear lower stresses. The imperfection of the actual tip enables the curve to closely align with the experimental data. Therefore, assuming that the tip has a smooth surface significantly influences the force required for indentation. This explains why spherical tips consistently exhibit a load underestimation.

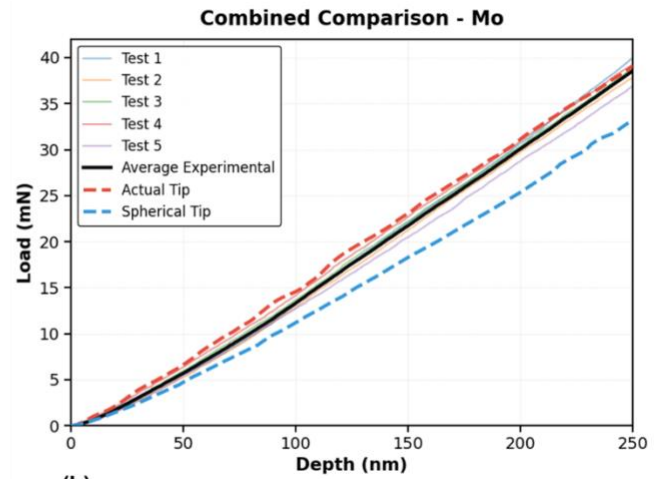
The relatively higher error observed in the cases of Mo and W is likely due to a combination of factors: the inherent variability in material properties, the assumption of isotropy in the analysis, and increased compliance-related errors during indentation experiments. However, it is difficult to isolate the contribution of each factor at this stage.

Table 3 Simulation errors in FEM from the load–depth curves with **(a)** spherical tip (radius of 4.60) **(b)** the actual tip at depth of 250 nm

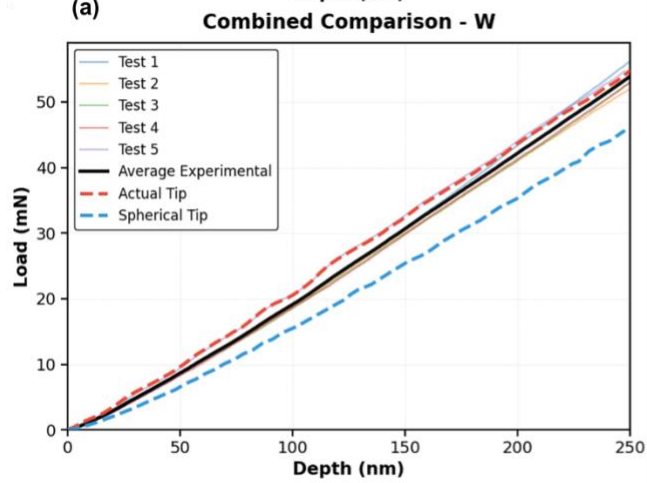
	Error (%)	Error (mN)	Experiment Error variation (%)
(a). Spherical Tip			
Cu	21.21	1.09	3.37
W	17.66	5.02	3.67
Mo	12.81	2.72	2.41
(b). Actual Tip			
Cu	1.15	0.07	3.37
W	3.32	0.77	3.67
Mo	6.90	1.07	2.41



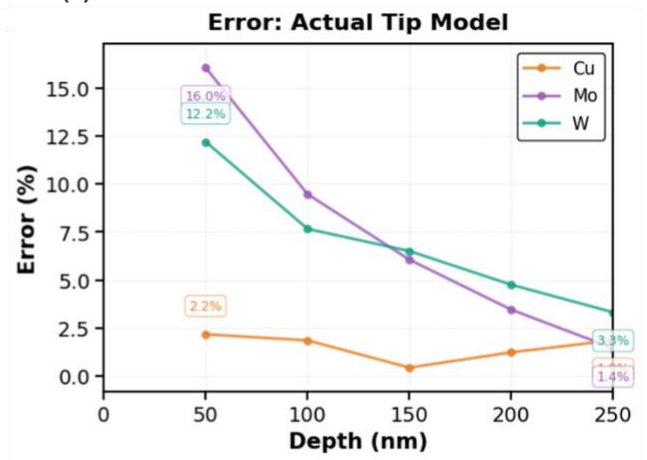
(a)



(b)



(c)



(d)

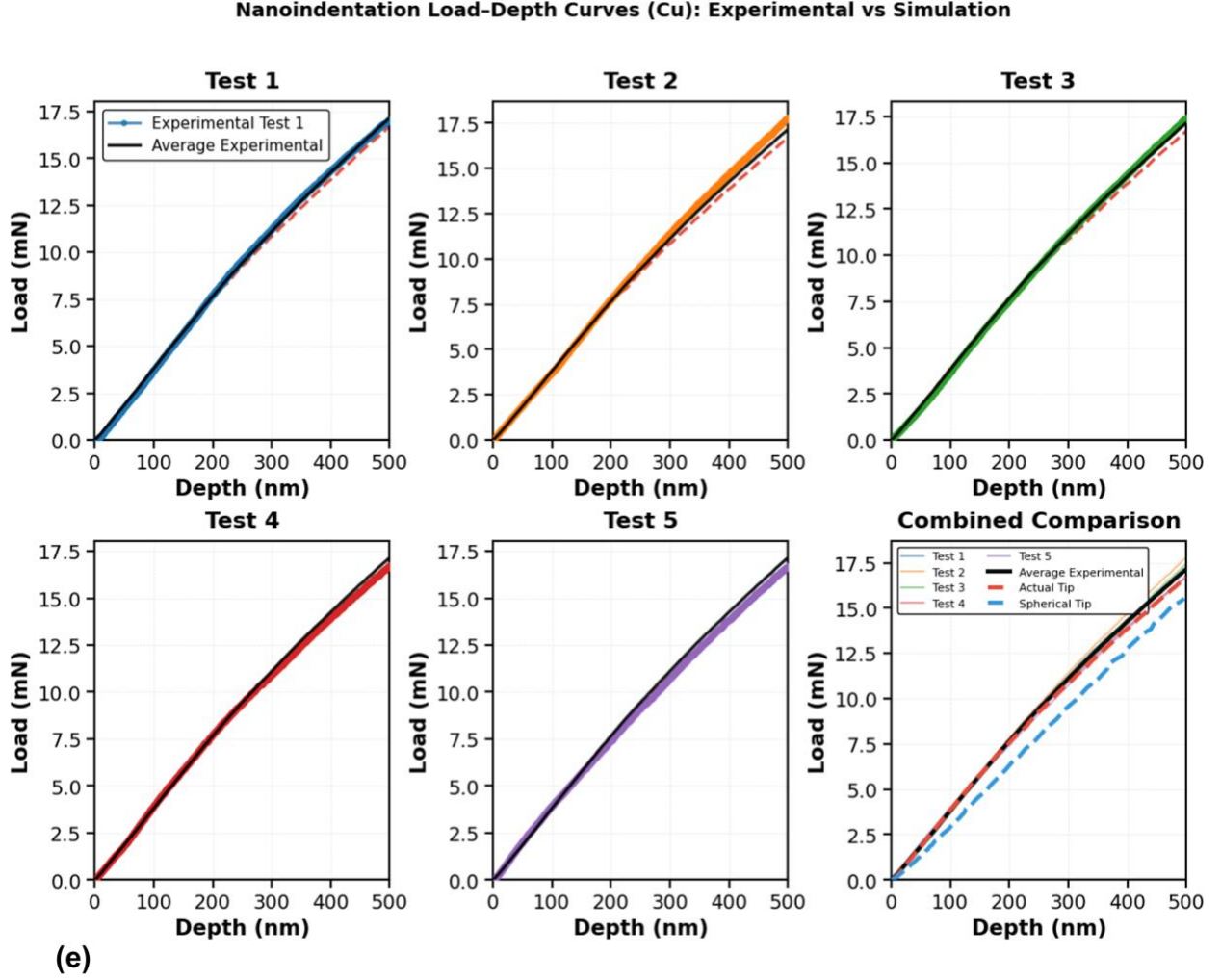


Figure 6. (a) Curves from simulation and experimental tests on copper (Cu) wire (b) Load–depth curves from simulation and experimental tests on molybdenum (Mo) wire, (c) Load–depth curves from simulation and experimental tests on tungsten (W) wire, (d) Maximum error as a function of indentation depth for Cu, Mo, and W, (e) Load–depth curves for Cu (to 500 nm).

4.2. Mesh study

Given that the true geometry of the indenter’s tip is essential for accurately modeling the spherical indentation at shallow depths, the next logical question to address is the mesh refinement necessary to obtain precise predictions not only of the load–depth curve but also the stress and strain associated with the experiments. A well-refined mesh can capture high stress and strain gradients, ensure that the representation accurately reflects the actual geometry of the indent and the material, and capture the intricate nonlinear behavior. Consequently, three types of meshes are studied: coarse, medium, and fine mesh. The details of these meshes are presented in **Table 2**. For

each model, the mesh becomes more concentrated towards the contact point of the actual indenter and the sample. The accuracy of tetrahedral meshes over other mesh shapes has been extensively studied and widely cited [58,59]. Although hexagonal shapes are generally preferred across literature (particularly in bone and organic structure studies), tetrahedral element meshes are often more practical for complex geometries. However, requiring graded mesh refinement around corners for polygonal and polyhedral domains is essential to optimize accuracy [58,59]. Guessasma and Bobzin advocated for the utilization of tetrahedral elements to enhance simulation accuracy. They employed 10-node tetrahedral elements for the indenter, while Bobzin utilized four-node linear tetrahedral elements for the material mesh [9,45]. **Table 2** provides comprehensive details of all the models defined in the FEM simulation. In the present work, these mesh configurations are therefore evaluated specifically for the experimentally calibrated spherical tip geometry and the range of properties considered and are not intended as universal mesh prescriptions for all indentation problems.

In addition to the mesh type, the computational time required for the analysis of each FEM simulation is a factor considered in this study. The temporal discrepancy between models is substantial, ranging from several minutes to nearly 24 hours. Numerous factors beyond mesh resolution contribute to computational duration, including indentation depth, system compatibility, modeling accuracy, and model complexity. Indentation depth significantly impacts the simulation time; for this investigation, which focuses specifically on mesh sensitivity, an indentation depth of 250 nm is selected to minimize computational costs. Accordingly, all timing and error comparisons reported here are restricted to this indentation depth and the loading conditions defined in this study.

It is well established that augmenting the mesh density within the models leads to a more accurate representation of load–depth curves compared to coarser alternatives. One “coarse” element can significantly disrupt the results for the entire simulation due to poor discretization [59]. Established literature demonstrates that the plastic deformation zone particularly requires denser meshes to yield accurate results. While the specifics of how the mesh density and size are selected are not well documented, it is generally understood that the appropriate mesh type and density should be determined based on practical considerations such as noise, accuracy in initial testing, and time efficiency [13,56,58,59]. Given the lack of mesh quality validation even in recent literature, this research evaluates the appropriate mesh based on accuracy and relative time

consumption for the present indentation geometry, material space, and depth range. Statistically, a coarse mesh exhibits an error margin of 2.5778 percent, while a medium mesh registers an error of approximately 0.8564 percent more in comparison to a fine mesh when reproducing the fine-mesh load–depth response for representative cases across the studied material property range. This indicates a marginal discrepancy of about 1 percent between the medium and fine meshes. Figure 7 illustrates that significantly enhancing mesh resolution primarily mitigates the undulatory characteristics observed in the load–depth curve, which are pronounced in coarse meshes and moderately present in medium meshes. Furthermore, the ratio of elements relative to the distance to the plastic zone can be controlled according to general ratios such as less than 1:3 for optimal efficiency without compromising accuracy [56] in the class of spherical indentation models examined here. From a cost-effectiveness standpoint, the medium mesh is superior to the fine mesh, requiring approximately 1.5-2 hours to execute a model, as opposed to the extensive duration of 24 hours for the fine mesh while still capturing approximately 99 percent of the accuracy achieved with fine mesh resolution. However, it is important to note that applying a fine mesh to the W wire reduces the mean error percentage from 7.58% to a mere 3.32%, indicating that higher-strength materials can demand additional refinement when tighter error tolerances are required.

In conclusion, employing a medium mesh with specimen element type C3D10 (tetrahedral shape and quadratic geometric order) for the specimen, and indenter element type R3D3 (triangular shape), yield results that are approximately 99% representative of a fine mesh for the present spherical indentation models and material property range. In this study, the medium mesh is therefore adopted as the default mesh resolution for large-scale simulations, as it provides an acceptable balance between accuracy and computational cost. This approach is also more cost-effective, as the model requires less than two hours to analyze. Alternatively, if accuracy is critical to minimizing the overall error percentage, using a fine mesh is the superior option, as the error percentage tends to increase with alloys of higher strength, such as W. Importantly, these conclusions are problem-specific: for any new FEM calculation model with different material behavior, indenter size, geometry, or target output quantity, a dedicated mesh-convergence verification must be performed to confirm that the chosen grid density is adequate.

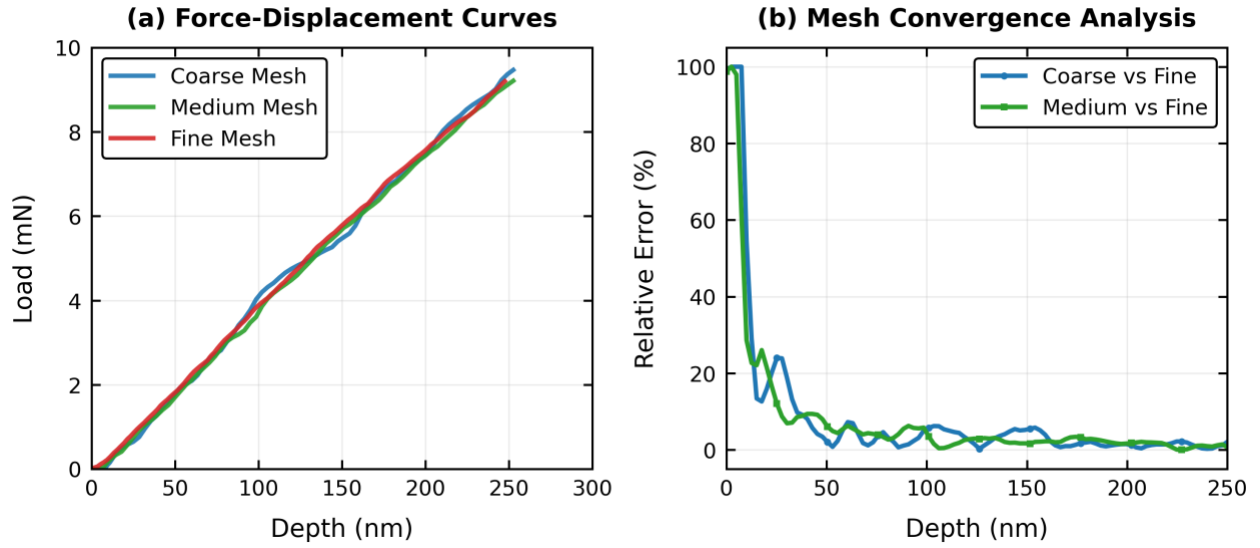


Figure 7. (a) Load–depth curves and (b) Mesh convergent analysis.

4.3. Friction effect of actual tip

In investigating the effects of friction between the indenter and the specimen, several simulations with varying friction coefficients were analyzed using medium mesh resolution. To identify the impact of friction, seven models employing Cu wire with actual tips were tested. These tests considered frictional contact at the indenter interface using friction coefficients of 0.2, 0.3, 0.4, 1, and 1.5, as depicted in Figure 8. This graph illustrates nanoindentation load–depth simulation results for Cu wire using various friction coefficients.

Comparing these models to empirical test averages, simulations with a friction coefficient of 0.4 exhibited the closest association, demonstrating an error margin of only 1.47 percent. This represents approximately a 0.4 percent improvement over frictionless scenarios and about a 0.6 percent enhancement compared to individual experimental averages. This finding is significant; it implies that a modification in friction can modestly yet effectively enhance simulation accuracy, achieving greater fidelity. However, given that the error between the simulation and the experimental average is less than the inherent variations between the tests, it is not possible to evaluate the ‘true’ coefficient of friction representative of the contact between the indenter and the sample in practical application scenarios.

Figure 8 suggests that variations in friction have a minimal influence on simulation outcomes, as their impact on error percentage remains negligible. While the coefficient of friction does play a role in simulation accuracy, its effect is relatively small, particularly at the nanoscale. This behavior is consistent with prior finite-element studies of static spherical indentation, which report that the load–depth response is only weakly sensitive to interfacial friction when the contact area is small and penetration depths are shallow [60,61]. As shown in **Table 4**, the force error decreases by only 0.07 mN when the total load is in the range of several mN. While this reduction may be relevant when using load–depth curves to predict elastoplastic material properties [46,47], its influence on overall error remains limited, thereby minimizing any confounding effects when analyzing newly discovered materials.

On the other hand, in Figure 8(b), there is a strong dissociation between the forces calculated from a simulated ideal spherical indenter and the forces obtained in real experiments. Figure 8 demonstrates the visual disparity and weak correlation between unadjusted ideal spherical indentation simulation and simulation using an actual tip. To be precise, there is a marked reduction of error by 20.33% from assuming a frictionless contact. The surface of the indenter is sufficiently smooth that it does not manifest itself in the load–depth curve, but frictional effects may become noticeable at larger indentation depths.

The reduction in error arises from changes in tip geometry and the interactions between the indenter and the sample surface. The difference in geometry holistically affects surface roughness (at the nanoscale), depth, and the contact area (contact interactions in general), which are crucial to calculating the mean load for a given depth. Given the relatively high error exhibited by the spherical indenter, the change in friction appears to have a less significant impact; as illustrated in **Table 4**, it is evident that assuming a coefficient of friction of 0.4 yields the least amount of error in our results for the actual tip. Nevertheless, this does not imply that a coefficient of friction of 0.4 is necessarily the optimal value for experiments.

Several additional simulations are performed with Mo and W, and it is observed that the optimal outcome is achieved with a friction coefficient of one; however, this effect is relatively low. We notice that the error percentage for Mo has decreased from 4.89% to 4.47%, and for W, it has decreased from 7.46% to 6.82%, including the friction coefficient of one.

The prevailing assumption is that friction coefficients do not substantially influence the load–depth curve. This assumption is valid primarily for shallow penetration depths and small contact areas, as supported by prior numerical studies, and does not imply that friction can be neglected for deeper indentation or large-contact regimes. An increase in the friction coefficient may enhance the material’s hardening or softening behavior, thereby affecting its resistance to sliding and necessitating a different applied load to achieve the same depth. **Table 4** demonstrates that by increasing the friction coefficient to 0.4, there is a slight reduction in error compared to the experimental results when modeling with the actual tip. Nevertheless, at a coefficient of 1.5, the results deteriorate substantially, suggesting that excessive friction may lead to an overestimation of material stiffness.

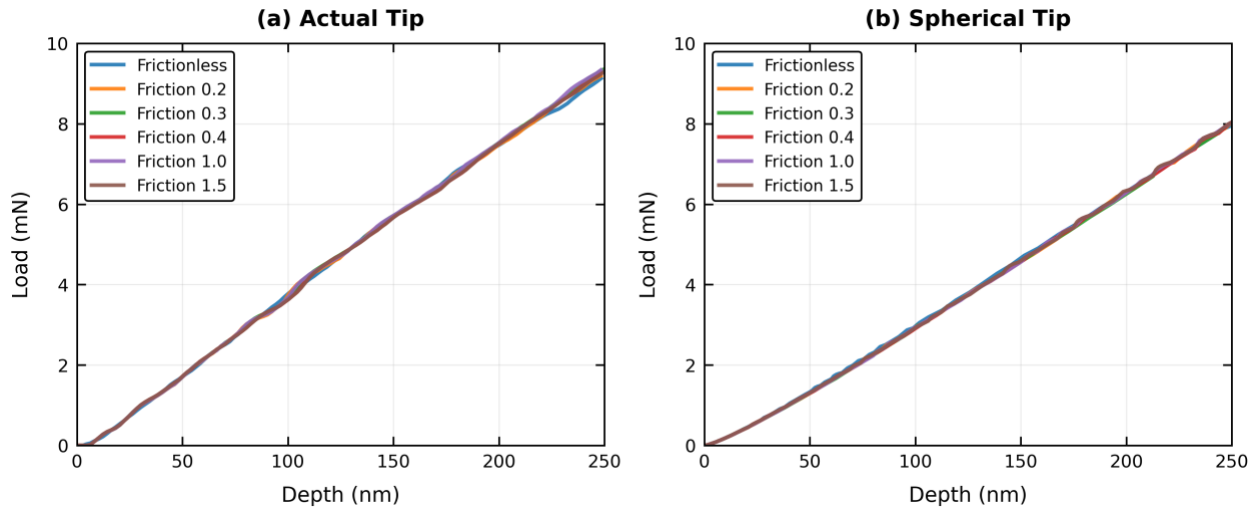


Figure 8. (a) Load–depth curve of the actual tip model due to the influence of different coefficients from frictionless conditions, **(b)** Load–depth curve of the spherical indenter model due to the influence of different coefficients from frictionless conditions.

Table 4 Comparison of identification results: the effect of friction in actual and spherical (FEM) vs average nanoindentation tests on Cu.

Type of Simulation	Friction Coefficient	Error Mean (%)	Force Error (mN)
Actual Tip	Frictionless	2.07	0.09
	0.2	1.77	0.07

	0.3	1.75	0.06
	0.4	1.47	0.05
	1	1.89	0.07
	1.5	1.49	0.05
Spherical Tip	Frictionless	20.33	1.13
	0.2	21.26	1.17
	0.3	21.40	1.18
	0.4	21.26	1.17
	1	21.14	1.16
	1.5	20.92	1.15

4.4. The effect of specimen's boundary condition in FEM nanoindentation with actual tip

These high-fidelity true tip geometry simulations require a greater amount of computational time compared to simulations using idealized geometry. Also, multiple nanoindentation simulations are frequently employed to comprehend various physical mechanisms, such as pile-up or sink-in effects observed in experiments. To mitigate the increased computational time, we investigate the minimum sample dimensions necessary to accurately predict the load distribution from simulations. We employ the actual tip geometry model and vary the dimensions of the specimens being indented to determine the minimum dimensions required to faithfully capture the indentation response. It is important to note that the primary objective of this section is not to identify a criterion for minimum sample dimensions but rather to comprehend the influence of various factors on the minimum dimensions required for analysis.

To assess the impact of specimen size, seven tests were modeled using different sizes ranging from 9.2 μm to 50 μm . From an experimental perspective, the true diameter of the Cu wire is around 45 μm . Therefore, the results are compared with the load–depth curve obtained with diameter (D) = 45 μm . It is noteworthy that any significant differences between $D = 45 \mu\text{m}$ and $D = 50 \mu\text{m}$ would indicate substantial compliance-related errors in the experiment. However, as evidenced by **Table 3**, there exists a minimum error in the resulting corrected and original load–depth curves. Each model exhibited consistency, with variations primarily attributed to sample

dimensions. To ensure cost-effectiveness while maintaining accuracy, the medium mesh resolution was employed for each case.

Figure 9 illustrates the impact of altering the sample diameter on the load–depth curve. Notably, there appears to be no significant deviation in the load–depth curve until a depth of 250 nm. For diameters exceeding 30 μm , the error between simulations and the specimen size of 45 μm is less than 0.79%. Overall, within the tested diameter range (9.2–50 μm), the differences in the load–depth response remain below 1%. Conversely, for specimen diameters smaller than 10 μm , the error in the load–depth curves become notably higher. Specifically, for a diameter of 9.2 μm , the error is approximately 3.40%. In addition, the stress field for $D = 9.2 \mu\text{m}$ indicates that size-constraint effects become more pronounced at this finer scale, suggesting that further reductions in specimen diameter would likely increase the error beyond the values reported here. Therefore, in the case of Cu, the influence of sample dimensions appears to be minimal.

Figure 9(b) and **Figure 9(c)** depict the von Mises stress distribution under the indenter for a specimen with a diameter of 9.2 μm and a specimen with a diameter of 45 μm , respectively. As is evident, the von Mises stress marginally increases from 0.4468 GPa to 0.4670 GPa, indicating that smaller specimens experience a more pronounced strain gradient near the indenter tip. This gradient leads to higher local stresses, as materials exhibit enhanced resistance to deformation when their volume is restricted. Although the difference between these two values is relatively small, it suggests a trend where further reduction in specimen size may lead to increased stress due to the intensified strain gradient.

Consequently, it can be inferred that reducing the size of the model domain to a smaller diameter generates stress concentrations at the boundary condition layer, resulting in an uneven stress distribution. This phenomenon implies that the elements and material cannot deform as freely as they would in an infinite medium. Consequently, altered stress distribution occurs, leading to apparent changes in stiffness or softness. These variations are influenced by both the material type and boundary conditions, such as clamping effects or proximity to a hard surface. Therefore, for the unit work done while increasing the depth, the energy dissipated in plastic deformation is more pronounced in the constrained case of 9.2 μm , resulting in lower load (and reaction force) for the given depth.

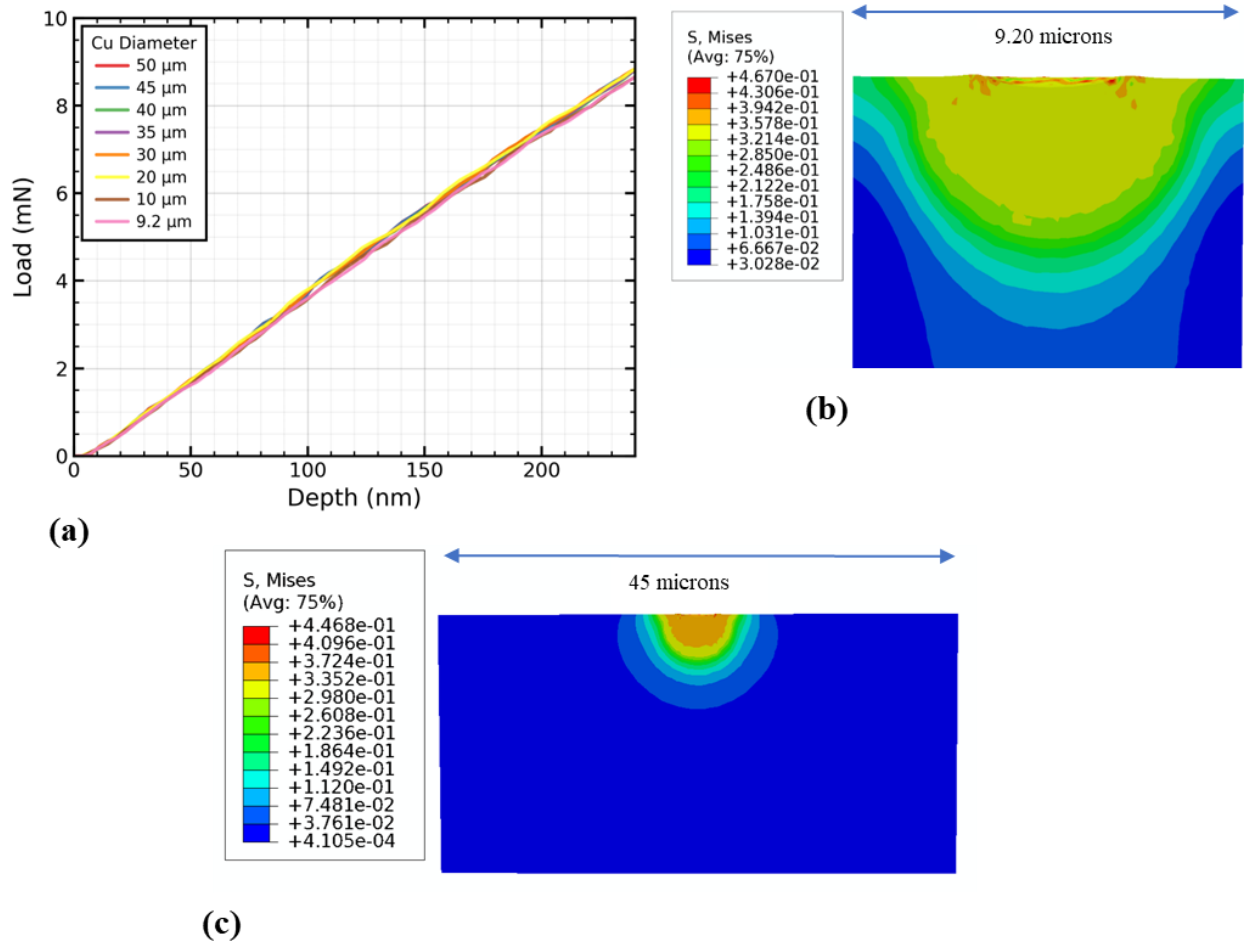


Figure 9. (a) Load–depth curve of the effect of different specimen’s size effect, (b) Specimen sample: cross-sectional diameter 9.2 micrometers, depth 25 micrometers, (c) Specimen sample: cross-sectional diameter 45 micrometers, depth 25 micrometers.

4.5. Investigating influence of indenter radius

A central practical question in instrumented indentation is whether a simple 2D axisymmetric model with a single spherical radius can reproduce the load–depth response obtained with a fully resolved, non-ideal tip geometry. This question matters because a scanned tip gives the highest fidelity but requires extra experimental effort and computational cost, whereas a 2D spherical indenter is inexpensive and fast. We addressed this by comparing several candidate spherical radii against both experiments and a 3D actual-tip model for Cu, Mo, and W over depths up to 250 nm, with primary attention on 100 to 200 nm where tip rounding is most influential and where future thin-film work will ultimately operate to avoid substrate effects.

Across the tested radii, a value of 5.80 μm , about a 26 percent increase over the nominal 4.60 μm , produced the lowest average error in the range of 250 nm window for all three materials. In that window, the calibrated spherical curves closely track both the experiments and the 3D actual-tip curves, with mean accuracy near 96.5 percent as summarized in **Table 3 and Figure 10**. **This shows** that an optimal spherical radius can exist for a specified depth window and for a small class of materials with differing properties.

Meanwhile, the use of a 2D simulation has limitations. By the 250 nm depth, the 5.80 μm curves begin to diverge from both the actual-tip and experimental curves for all three materials, and this divergence is expected to grow at greater depths if the radius is held fixed. A single radius therefore cannot match the entire 0 to 500 nm range across Cu, Mo, and W. The 3D scan also shows slight departures from a perfect sphere, which introduce small azimuthal variations in stresses and plasticity that an axisymmetric sphere cannot reproduce.

The calibrated radius is consistent with expected mechanical behavior. A larger sphere engages a larger contact area and a greater material volume at the same depth, which raises the required load, in line with Hertzian scaling and the deeper, wider plastic zones observed in our simulations. Within 100 to 200 nm the von Mises fields for the 5.80 μm sphere and the actual tip are very similar, which explains the close agreement in load–depth response. At larger depths, differences in evolving contact shape and area amplify small geometric discrepancies and break that

These findings support a nuanced approach to modeling practices. When a scanned tip profile is available and computational resources permit, the actual tip geometry should be employed, particularly in analyses involving a wide depth range, sensitivity to directional stress fields, or the need to resolve subtle geometry-induced effects. In scenarios where tip scanning is unavailable or computational cost is prohibitive, a calibrated spherical tip radius offers a practical alternative within a constrained depth window. This radius should be empirically calibrated using one or more benchmark materials representative of the intended depth range and subsequently applied in 2D axisymmetric simulations restricted to that window. The calibration must be treated as specific to the tip and experimental setup and should be revalidated if the material system or depth range changes. Extrapolation of a single calibrated radius beyond its validated range is

discouraged. For broader depth coverage, consider implementing a depth-dependent effective radius or reverting to the scanned tip geometry.

The identification of a practical optimum within the 0-200 nm depth range does not imply the existence of a universal tip radius applicable across all depths or material systems. Rather, it reflects a localized equilibrium among tip rounding, contact area evolution, and material-specific mechanical response. The deviation observed at 250 nm underscores the limitations of assuming a single radius can adequately represent all conditions and cautions against overinterpreting a close fit within a narrow depth band as evidence that tip geometry can be fully characterized by a single parameter.

A substantial portion of the literature employs calibrated or modified tip geometries fitted to reference materials. Our findings contribute to this discourse by quantitatively delineating both the strengths and limitations of such an approach. While a single calibrated radius can yield high fidelity for a small class of metallic systems within a defined depth window, accuracy degrades as depth increases. This trend provides insight into the frequent dataset- and window-specific nature of reported calibrations.

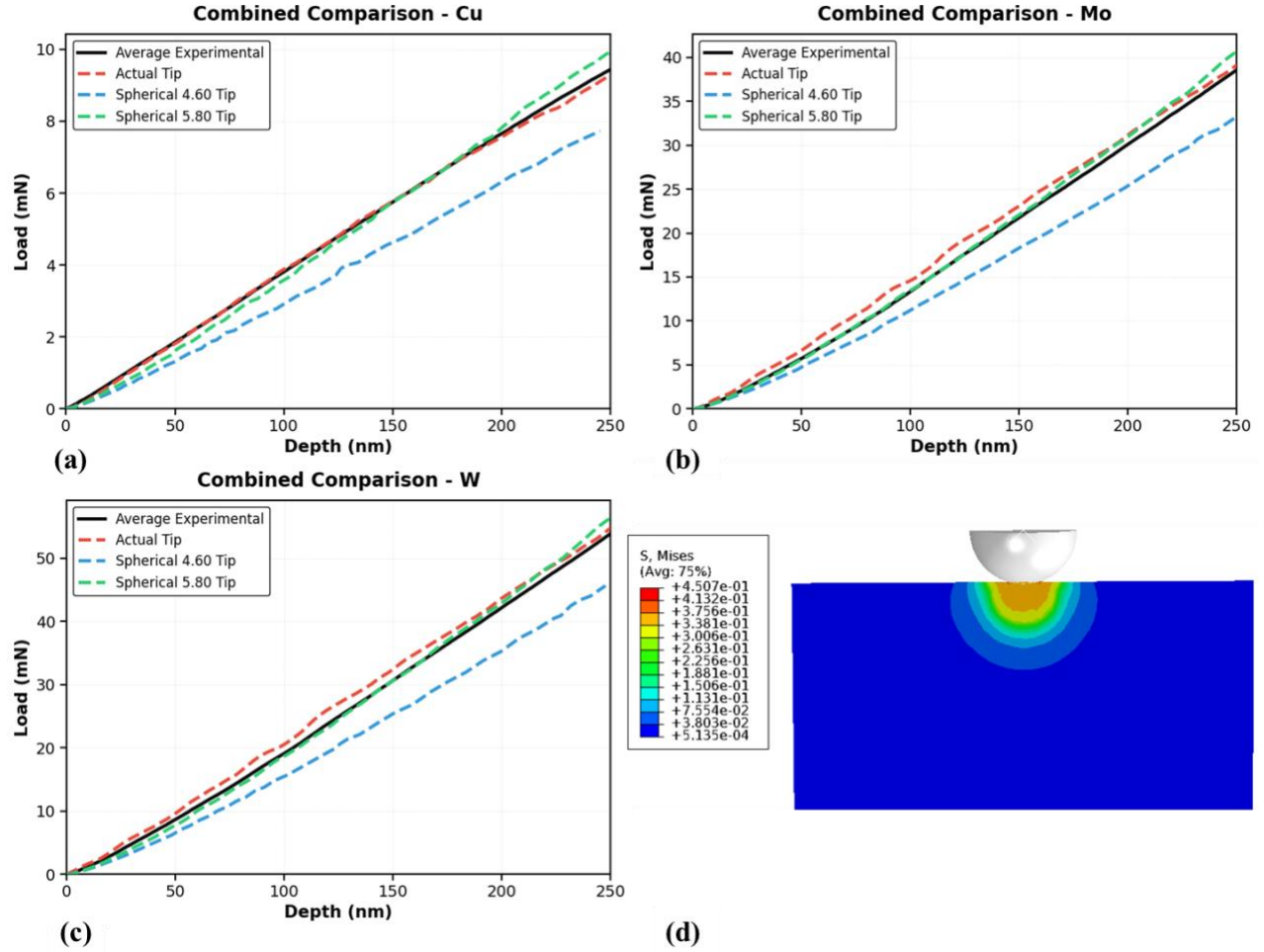


Figure 10. (a) Load–depth curve of the increasing size of the spherical indenter on copper (Cu) wire (b) Load–depth curve of the increasing size of the spherical indenter on tungsten (W) wire, (c) Load–depth curve of the increasing size of the spherical indenter on molybdenum (Mo) wire, (d) Von Mises Stress in Cu Wire.

4.6. Overall findings

It is evident that employing the actual tip significantly reduces the error percentage from more than 20% to as low as 1%. In contrast to methods that are material or test environment dependent, the current study offers several advantages once the tip geometry has been scanned. The results show that incorporating the actual scanned tip geometry most strongly reduces error in the load–depth response, whereas variations in friction is secondary within the range tested. Perhaps as the depth is increased, frictional effects should be taken into account; however, the current study is

limited to shallower depths. Practically, the scanned tip is preferred when accuracy across a broad depth window is required, when directional stress fields matter, or when extending to thin films where shallow depths are mandatory; a calibrated spherical radius can be acceptable within a narrowly defined depth window when cost or scanning access is limited. Key considerations include scan longevity as the tip wears or chips and the breadth of materials over which a single scan remains valid. In our study, the actual-tip approach remains accurate through at least 500 nm and is expected to continue to perform well at greater depths within the tested material set. Future work should track scan validity over time, test transferability across a wider set of materials, and formalize depth-dependent calibration to bridge the gap between calibrated spheres and full tip geometry. The present conclusions are restricted to shallow indentation depths where rate-dependent and size-dependent effects are expected to remain secondary.

Conclusion

This study established a high-fidelity forward finite element modeling framework for shallow spherical nanoindentation by directly incorporating the measured three-dimensional geometry of the indenter tip. Across Cu, Mo, and W wires with distinct elastic-plastic behavior, the use of the actual scanned tip substantially improved agreement between simulation and experiment relative to nominal-radius spherical models, reducing load-depth error from more than 20% to as low as about 1% in the best case, while also providing consistently improved predictions across the tested material set. These findings confirm that accurate representation of the true spherical tip geometry is a dominant requirement for reproducing shallow indentation responses, particularly in regimes where the load-depth curve is highly sensitive to contact geometry and early plastic deformation.

Beyond demonstrating improved agreement for a specific set of experiments, the present work clarifies the broader role of forward-model fidelity in spherical nanoindentation. Because the load-depth curve serves as the primary observable for subsequent inverse analysis, errors introduced by idealized tip geometry can distort the relationship between measured indentation data and the underlying constitutive behavior. By establishing a geometry-faithful forward model, the present framework provides a physically grounded and transferable basis for future inverse extraction of stress-strain behavior, including machine-learning-assisted workflows aimed at high-throughput

characterization of thin-film and combinatorial material libraries. In this sense, the current study is intended not as a complete inverse solution, but as a necessary first step toward such a workflow.

Additional parametric studies further clarified the relative importance of other modeling choices within the shallow-depth regime examined here. Mesh refinement strongly influenced numerical smoothness and accuracy, with the medium mesh providing a practical balance between computational cost and predictive fidelity for most cases, while higher-strength materials such as W benefited from finer discretization. Friction had only a secondary influence on the load-depth response within the penetration depths considered, indicating that geometry dominates the model-experiment discrepancy in this regime. A calibrated spherical radius was shown to provide a useful lower-cost approximation over a limited depth window, but it could not fully reproduce the broader response captured by the scanned three-dimensional tip geometry. These findings suggest that calibrated-radius approaches may be acceptable for restricted applications, whereas direct use of the measured tip is preferred when broad depth validity, directional stress fidelity, or extension to shallow thin-film problems is required.

The conclusions of the present study are restricted to shallow indentation depths and the loading conditions considered here, where rate-dependent and indentation size effects are expected to remain secondary relative to the geometric effects under investigation. Future work should examine the extent to which the measured tip remains valid over longer experimental use, assess transferability across a broader range of materials and film-substrate systems, and integrate the present geometry-consistent forward framework into inverse and machine-learning-based pipelines for rapid extraction of constitutive behavior from spherical nanoindentation data.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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